PROJECT CURRENT STATUS

CMS COLLEGE OF ENGINEERING — DIGITAL ADMISSION PORTAL

(AUTONOMOUS INSTITUTION — AFFILIATED TO ANNA UNIVERSITY & APPROVED BY AICTE)

CMS Nagar, Ernapuram Post, Near Collectorate, Namakkal - 637 003, Tamil Nadu

• SYSTEM STATUS: 100% OPERATIONAL & PRODUCTION LIVE

Project Title	**Project Current Status**	Report Version	v2.4.0 (Production Master)
Operational Condition	🟢 **ACTIVE & FULLY VERIFIED**	Report Compiled On	November 1, 2026
Live Production Web URL	**https://cms-admin-system-nvoj.onrender.com**		
Source Code Repository	https://github.com/cmsadsystem-ai/CMS-ADMIN-PORTAL (branch: `main`)		
Central Cloud Database	**TiDB Cloud Serverless MySQL** — Permanent Free Forever ($0/month, No 30-Day Expiry)		
Live Real-Time Engine	**Socket.IO WebSockets + ASGI** — Zero-Refresh Live Dashboard Synchronization		

1. Executive Summary

The **CMS College of Engineering Admission & Admin Portal** is a cloud-native, autonomous web management system engineered to automate student registration, calculate Anna University engineering cut-off marks in real-time, deliver synchronized administrative review dashboards for staff, dispatch official print-accurate paper application emails, and persist all admission records into a permanent central cloud database.

All temporary trial limitations (such as the 30-day Aiven MySQL trial) have been completely eliminated by deploying **TiDB Cloud Serverless MySQL**. The software requires $0.00/month to operate and is permanently stable for institutional deployment.

2. Hosting & Deployment Ecosystem (Render)

Hosting Architecture: Deployed as a single unified web service container on **Render Cloud (Singapore Region)** for ultra-low latency access in Tamil Nadu and South India.

- **Frontend Layer:** High-performance React SPA built with Vite, serving static distribution bundles.
- **Backend Layer:** Python 3.11 with FastAPI ASGI and Uvicorn server.
- **Automated Pipeline:** Any commit pushed to the GitHub `main` branch automatically triggers a zero-downtime production deployment via `render.yaml` Blueprint.

3. Central Cloud Database Status (Permanent Free TiDB MySQL)

Connected to **TiDB Cloud Serverless MySQL** (AWS Singapore Region). It provides 5 GB of permanent free relational storage with zero recurring fees and no trial expiration.

```
Host: gateway01.ap-southeast-1.prod.aws.tidbcloud.com
Port: 4000
Database: cms_admission
User: 2dqhC13C1WmSRUJ.root
Password: NNCSNv5IG4Gz2ipP
SSL Mode: Required (TLS enabled)
Connection URI: mysql+pymysql://2dqhC13C1WmSRUJ.root:NNCSNv5IG4Gz2ipP@gateway01.ap-southeast-1.prod.aws.tidbcloud.com:4000/cms_admission
```

Table Name	Primary Key	Purpose & Managed Fields
portal_applications	`id` (e.g. 260002)	Stores student admission applications, marks, status, timestamps, and full 100% form data in `data_json`.
staff_users	`id`	Manages administrative logins for System Admin, Admission Officer, and Vice Chairman (bcrypt passwords).
courses	`id` / `code`	Approved UG & PG engineering programmes (AI&DS, CSE, CSE-CS, BME, EEE, ECE, MECH, AI&ML, M.E).
fee_structures	`id`	Standard government & management fee rates categorized by community (OC, BC, BCM, MBC, SC, ST).
application_sequences	`year` (2026)	Atomic counter guaranteeing clean sequential application numbering formatted as `YYNNNN`.
portal_settings	`key`	JSON key-value storage for payment settings, campus backgrounds, and banners.

4. Real-Time Multi-User Synchronization Status (Socket.IO)

Zero-Refresh Real-Time Architecture: When any applicant or staff submits an admission form:

1. The application is atomically saved in the central TiDB Cloud database.
2. The server emits a real-time `application_created` or `application_updated` WebSocket event across `staff_channel` and `vc_channel`.
3. All open Admission Officer and Vice Chairman dashboards receive the event and **instantly add or update the record within 1 second — without anyone having to press F5 or refresh the page!**
4. **Optimistic Concurrency Control:** Conflicting simultaneous edits are intercepted with HTTP 409 conflict checks to guarantee data integrity.
5. **Auto-Reconciliation:** When a device's internet drops and reconnects, it automatically reconciles against TiDB Cloud.

5. Email Automation & Application Preview Status

Relay Engine	Google Gmail SMTP (TLS encrypted port 587)	**Sender Account**	`cmsadsystem@gmail.com`
Triggers	Dispatched on Status: **Enquiring** and **Confirmed**	**Recipient**	Applicant Email + CC to Admissions Desk

Authentic Digital Paper Application Format: The generated email body replicates the original physical CMS admission form:

- Official Autonomous College Header & 20 Years Educational Excellence Seal.
- Consultant & Organization identification block.
- Student & Parent profiles with formatted Aadhaar and contact numbers.
- Academic performance marks table (Maths, Physics, Chemistry, Cut-off, Total).
- 8-programme preference check grid with official Anna University course names.
- Approved tuition & hostel fee breakdown table.
- High-resolution digital signatures of Student and Parent.
- Office Use verification checklist (Original & Xerox certificates).
- Official signatory lines for **Admission Officer** and **Vice-Chairman**.
- **Refinement:** External portal view buttons have been cleanly removed for an authentic official document appearance.

6. Academic Formula & Cut-Off Computation

The system enforces Anna University's official engineering cutoff formula implemented in `app/cutoff.py`:

Cut-off = Mathematics + (Physics / 2) + (Chemistry / 2)

(Evaluated out of 200 marks, rounded to 2 decimal places)

7. Master Credentials Directory

A. Staff Web Portal Logins

Role	Username	Default Password	Access Level
System Administrator	`admin`	`change-this-password`	Master system setup, courses, fees, and staff accounts.
Admission Officer	`officer`	`CMSdemo1!`	Intake reviews, certificate verification, and status updates.
Vice-Chairman (VC)	`vc`	`CMSdemo2!`	Quota tracking, scholarship approval, and final admission authorization.

B. Infrastructure & Services Credentials

Service	Host / Resource	Username	Credentials / Secret
TiDB Cloud MySQL	`gateway01.ap-southeast-1.prod.aws.tidbcloud.com:4000`	`2dqhC13C1WmSRUJ.root`	`NNCSNv5IG4Gz2ipP`
Gmail SMTP Relay	`smtp.gmail.com:587`	`cmsadsystem@gmail.com`	`ohhl usqb hwso fjco` (App Password)
GitHub Repository	`cmsadsystem-ai / CMS-ADMIN-PORTAL`	`cmsadsystem-ai`	Personal Access Token / SSH

8. Cost & Lifetime Hosting Efficiency

Platform Component	Hosting Provider	Free Allocation Quota	Monthly Cost	Permanence Status
Full-Stack Application	Render Cloud	512 MB RAM / 0.1 CPU	**$0.00**	Active / Permanent Free
Relational Database	TiDB Serverless	5,000 MB (5 GB Storage)	**$0.00**	**Permanent Free Forever (No 30-Day Limit)**
Email SMTP Relay	Google Gmail	500 emails / day	**$0.00**	Permanent / Included
Source Control & CI/CD	GitHub	Unlimited Public/Private Repos	**$0.00**	Permanent / Included
TOTAL MONTHLY OPERATIONAL COST			**$0.00 / mo**	**100% Free Lifetime Operation**

9. Operational Runbooks & Live Verification

Runbook 1: Real-Time Terminal Cloud Database Watcher

To observe student registrations stream live in the terminal without opening any web browser:

```
cd "c:\College Projects\CMS-Admin-portal\cms-admission-backend\cms-admission-backend"
.venv\Scripts\python watch_live_db.py
```

Flashes automatically on the terminal whenever a new submission is written to TiDB Cloud.

Runbook 2: Querying Detailed Student Records in TiDB SQL Editor

In TiDB Cloud Console (`tidbcloud.com`) → `cms-admission-db` → **SQL Editor**:

```
USE cms_admission;

SELECT
    id AS `App ID`,
    student_name AS `Student Name`,
    status AS `Status`,
    course_name AS `Course`,
    JSON_UNQUOTE(JSON_EXTRACT(data_json, '$.studentMobile')) AS `Mobile`,
    JSON_UNQUOTE(JSON_EXTRACT(data_json, '$.fatherName')) AS `Father Name`,
    JSON_UNQUOTE(JSON_EXTRACT(data_json, '$.cutOffMarks')) AS `Cutoff`,
    JSON_UNQUOTE(JSON_EXTRACT(data_json, '$.mathematics')) AS `Maths`,
    JSON_UNQUOTE(JSON_EXTRACT(data_json, '$.physicsMarks')) AS `Physics`,
    JSON_UNQUOTE(JSON_EXTRACT(data_json, '$.chemistryMarks')) AS `Chemistry`,
    submitted_at AS `Submitted Time`
FROM portal_applications
ORDER BY submitted_at DESC;
```

10. Conclusion & Official Sign-Off

The **CMS College of Engineering Admission & Admin Portal** has achieved full technical and operational maturity. It eliminates physical paperwork delays, synchronizes admissions across administrative staff in real-time, guarantees zero data collision, and is backed by a robust, permanent free cloud database architecture.

SYSTEM ARCHITECT / DEVELOPER	ADMISSION OFFICER	VICE-CHAIRMAN